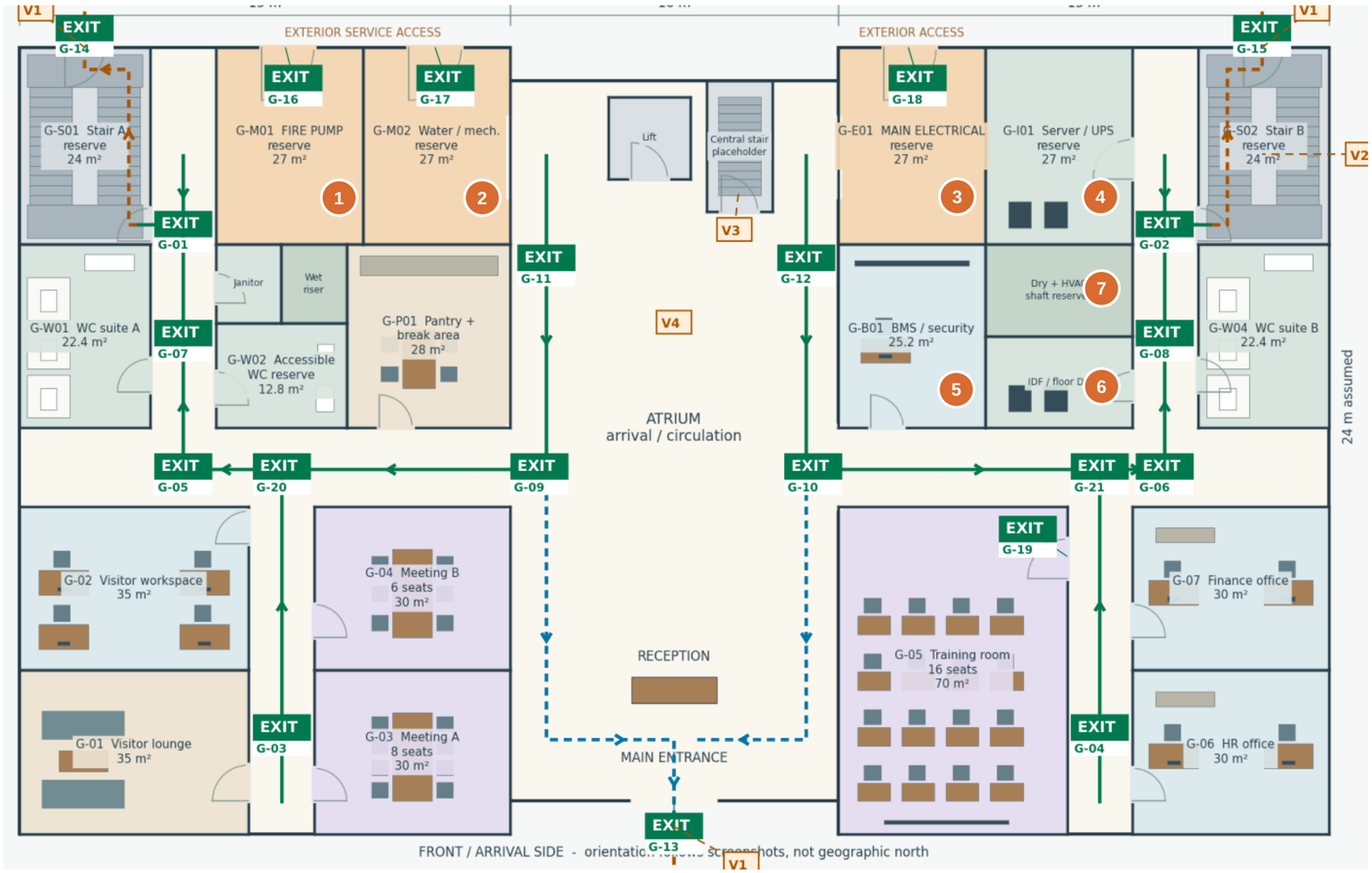
Ground floor | Populate the reserved rooms

Orange numbers identify rooms, not exact equipment footprints. Keep the existing walls, stairs and circulation routes; use your actual 3D dimensions.

PROPOSED REAR SERVICE YARD - outside the drawn building
Reserve tank zones near rooms 1 and 2; reserve a ventilated HVAC zone. Yard size and equipment separation are not established. Keep rear exits and service doors accessible.



1. **G-M01 / Fire pump room**
   Add FP-01 main pump, JP-01 jockey pump and separate controllers. Use the existing rear exterior door. Keep pump servicing and controller access clear.

2. **G-M02 / Domestic water**
   Add a proposed two-pump booster skid, pressure vessel and control panel. Reserve pipe entry from a domestic tank or incoming supply.

3. **G-E01 / Main electrical**
   Place MDB-G-01 cabinets along a side wall, with doors facing the room. Preserve the rear entrance and a clear working aisle.

4. **G-I01 / Server and UPS**
   Add a server rack, UPS and dedicated cooling. Keep rack front/rear access. Battery arrangement remains a separate design decision.

5. **G-B01 / BMS and security**
   Use the desk/display area for the operator station. Add a control/gateway cabinet on an accessible wall.

6. **IDF / floor distribution**
   Show an IT rack and DB-G as distinct equipment zones. The shared reservation needs partitioning/separation review.

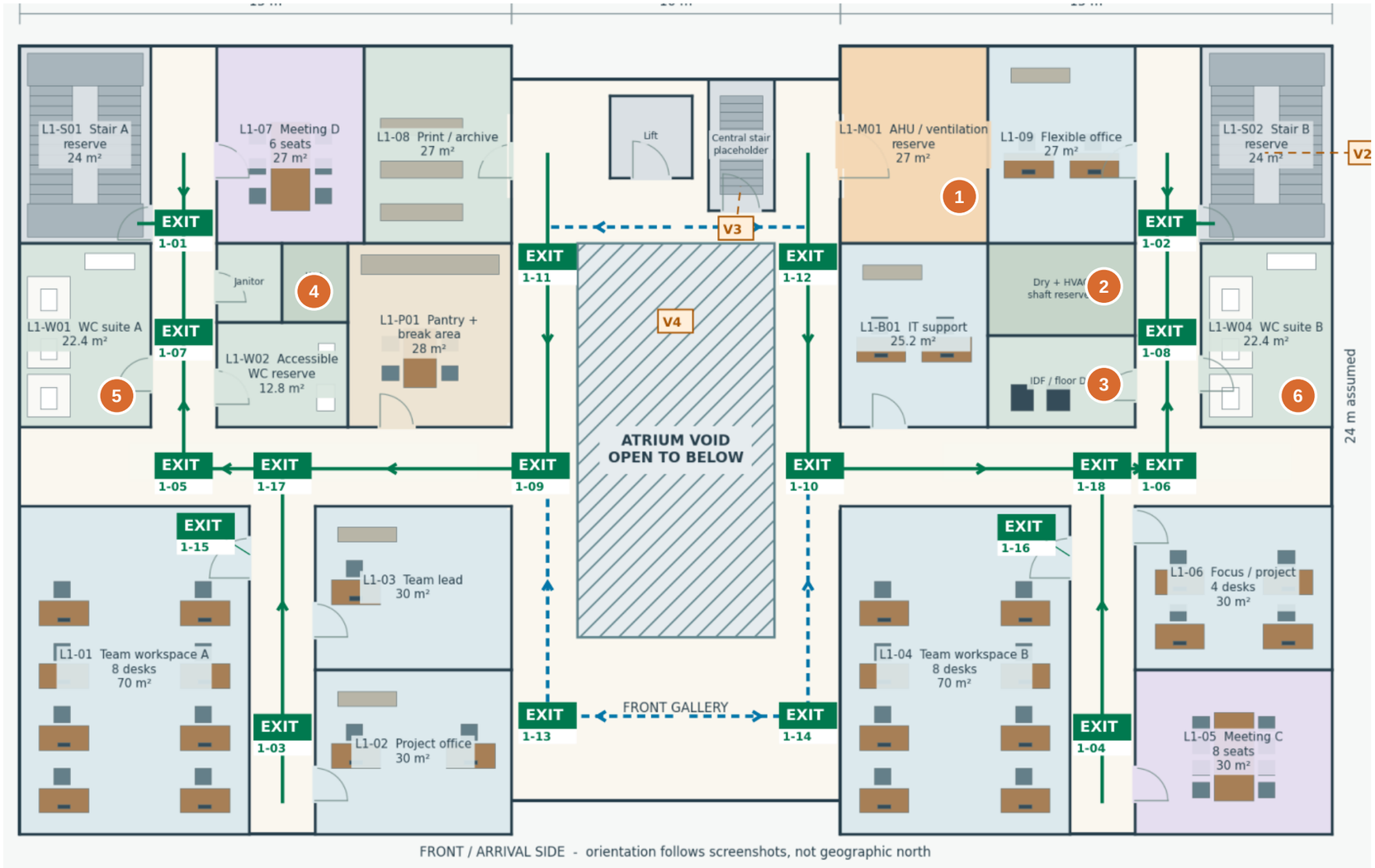
7. **Dry + HVAC shafts**
   Keep this space available for separate vertical service compartments. Do not use it for storage or a large floor-standing unit.

Start here: model rooms 1-4 first. Use movable equipment groups and separate clearance boxes. Tanks and fire pumps remain proposed until supply/demand calculations establish need and size. The source plan assumes a 40 x 24 m envelope; do not use it to rescale your existing model.

First floor | Ventilation and vertical services

Use the existing rear-right AHU room. Keep the atrium open and the occupied offices intact. The numbered locations repeat between upper floors.

REAR OF BUILDING
AHU room access needs to be resolved: a clear access door is not visible in the source plan. A west-wall door from the adjacent circulation is a candidate, subject to enclosure and egress coordination.



1. **L1-M01 / AHU room**
   Place AHU-L1-01 with its long axis front-to-back. Face service panels toward an open aisle. Reserve filter/coil removal space and an equipment replacement route.

2. **Dry + HVAC shafts**
   Use the HVAC compartment for vertical duct/refrigerant routes. Connect the AHU across a coordinated high-level service zone; do not cut through the adjoining office.

3. **IDF / floor distribution**
   Add DB-L1 and the floor network rack in separately coordinated zones. Stack service routes with the ground floor.

4. **Wet riser / janitor zone**
   Keep the existing left-side wet-service position. Reserve plumbing and drainage here. Do not place an AHU or electrical cabinet in this compartment.

5. **Left washroom group**
   Reserve an accessible extract-fan location and a dedicated exhaust path. Final fan count, duct size and discharge position remain to be designed.

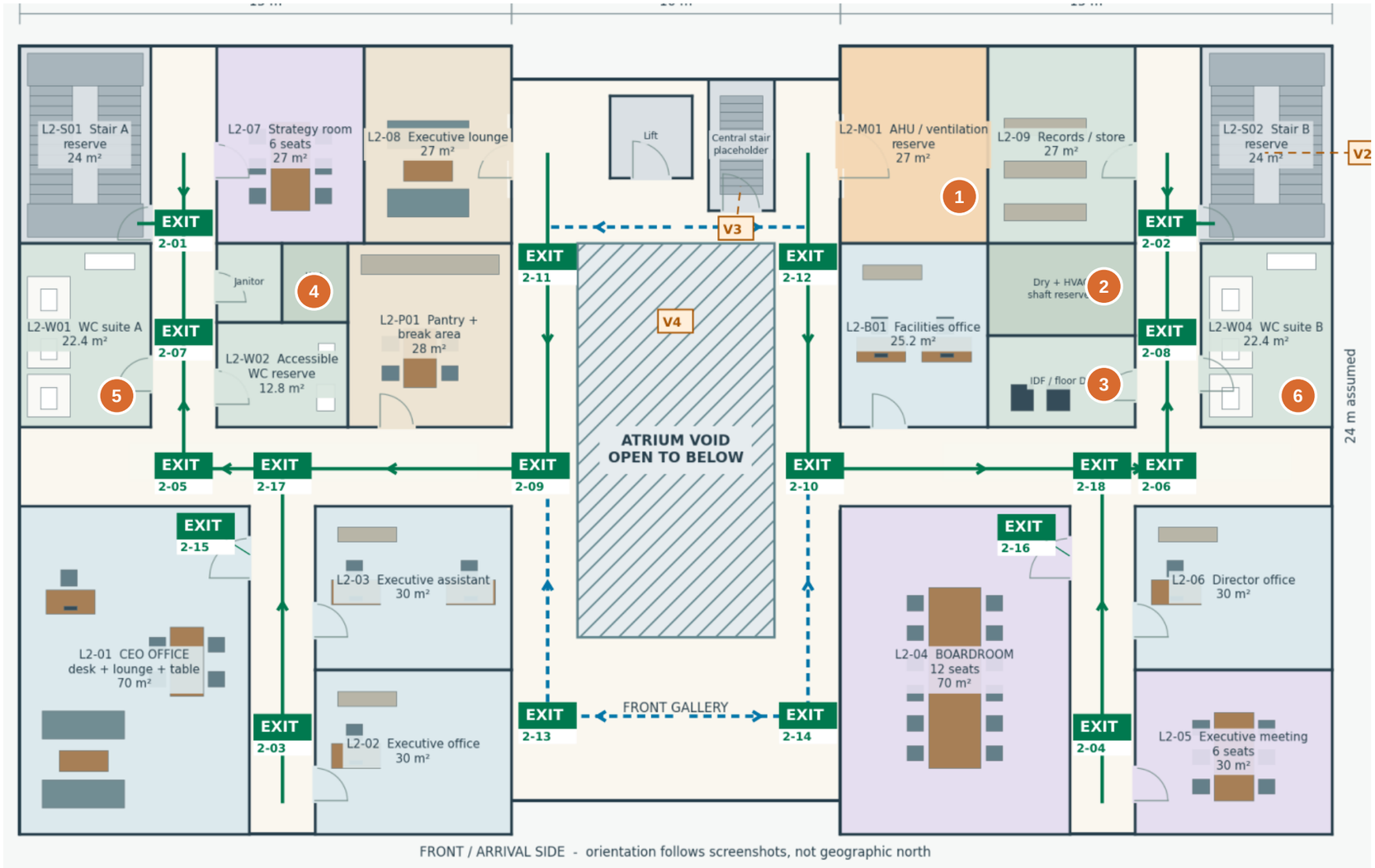
6. **Right washroom group**
   Provide a separate coordinated extract route for this side. Keep toilet exhaust separate from office return air in this concept.

Next modelling layer: route main services above corridors, then branch into rooms. Check beams, ceiling height, access and penetrations before drawing ducts. Outdoor condensers remain in the proposed ground-level yard; the atrium glass roof is not a plant platform.

Second floor | Ventilation and vertical services

Use the existing rear-right AHU room. Keep the atrium open and the occupied offices intact. The numbered locations repeat between upper floors.

REAR OF BUILDING
AHU room access needs to be resolved: a clear access door is not visible in the source plan. A west-wall door from the adjacent circulation is a candidate, subject to enclosure and egress coordination.



1. **L2-M01 / AHU room**
   Place AHU-L2-01 with its long axis front-to-back. Face service panels toward an open aisle. Reserve filter/coil removal space and an equipment replacement route.

2. **Dry + HVAC shafts**
   Use the HVAC compartment for vertical duct/refrigerant routes. Connect the AHU across a coordinated high-level service zone; do not cut through the adjoining office.

3. **IDF / floor distribution**
   Add DB-L2 and the floor network rack in separately coordinated zones. Stack service routes with the ground floor.

4. **Wet riser / janitor zone**
   Keep the existing left-side wet-service position. Reserve plumbing and drainage here. Do not place an AHU or electrical cabinet in this compartment.

5. **Left washroom group**
   Reserve an accessible extract-fan location and a dedicated exhaust path. Final fan count, duct size and discharge position remain to be designed.

6. **Right washroom group**
   Provide a separate coordinated extract route for this side. Keep toilet exhaust separate from office return air in this concept.

Next modelling layer: route main services above corridors, then branch into rooms. Check beams, ceiling height, access and penetrations before drawing ducts. Outdoor condensers remain in the proposed ground-level yard; the atrium glass roof is not a plant platform.

How to place each object in your 3D scene

Use the floor-plan markers for room selection, then use this checklist for the actual modelling. Suggested arrangements below are new proposals, not measured equipment layouts.

01 / Prepare the rooms

Hide the roof and upper floors. Confirm the room IDs and the FRONT/REAR direction against pages 1-3. Use your real model scale; the PDF geometry is assumed.

02 / Place simple objects

Start with rectangular equipment envelopes. Keep every pump, controller, board and AHU selectable. Place their bases on the room floor and their access faces toward a usable aisle.

03 / Preserve service access

Add a separate translucent object for each working/maintenance zone. Check doors can open and equipment can be removed. Do not treat a fixed generic clearance as code approval.

04 / Mark connection points

Give each asset labelled ports: pipe suction/discharge, duct inlet/outlet, electrical entry or condensate drain. Reserve routes before modelling detailed pipes and ducts.

05 / Add realistic details

After the envelopes fit, add pump/motor shapes, cabinet doors, AHU panels, grilles and equipment labels. Check the result in a person-height view as well as a top view.

06 / Prepare AURA selection

Use stable object IDs. Store room, floor, system and data source. Keep status values labelled SIMULATED until connected. Fire-system integration is monitoring-only for this prototype.

Illustrative object envelopes - metres, L x D x H; maintenance space excluded

FP-01 / main pump + motor	**2.0 x 0.9 x 1.1**	MDB-G-01 / switchboard	**2.4 x 0.8 x 2.0**
JP-01 / jockey pump	**0.5 x 0.5 x 1.0**	Server rack	**0.6 x 1.0 x 2.0**
Main fire-pump controller	**0.8 x 0.4 x 1.8**	UPS-G-01	**0.6 x 0.8 x 1.2**
Domestic booster skid	**1.5 x 0.8 x 1.2**	AHU / each floor placeholder	**3.0 x 1.2 x 1.5**

Do not size by appearance: these dimensions are modelling placeholders from the discussion, not selected products. Tank capacity, pump duty, electrical ratings, AHU air volume, condenser count and actual service clearances need design and manufacturer data.

Still unresolved: external yard dimensions; fire-water supply strategy; upper AHU door/replacement routes; floor/ceiling heights; fire-alarm panel and entrance annunciator location; atrium smoke strategy. The BMS room alone does not resolve fire-panel access. Retain the original egress and extinguisher information for separate coordination.